Kiyoung Om

Ph.D. Student, Electrical and Computer Engineering, The University of Texas at Austin

Machine Intelligence through Decision-making and Interaction (MIDI) Lab | Advisor: Prof. Amy Zhang

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Website | GitHub | LinkedIn | Google Scholar

RESEARCH INTERESTS

- Generative Models for Optimization and Control
- Alignment of Generative Models via Search and Fine-tuning
- Learning-based Approaches for Autonomous Vehicle Planning and Simulation

EDUCATION

The University of Texas at Austin | Aug 2026 - Present

Ph.D. in Electrical and Computer Engineering | Austin, TX, USA

Advisor: Prof. Amy Zhang, MIDI Lab

Korea Advanced Institute of Science and Technology (KAIST) | Mar 2024 - Feb 2026

M.S. in Data Science | Daejeon, South Korea

Advisor: Prof. Jinkyoo Park, System Intelligence Lab

GPA: 3.78/4.3

Korea University | Mar 2018 - Feb 2024

B.S. in Industrial Management Engineering | Seoul, South Korea

GPA: 4.02/4.5 | Major GPA: 4.19/4.5

Military Service: Republic of Korea Army, Sep 2019 - Feb 2021

PUBLICATIONS

* Equal contribution.

Journal Articles

Flow-ERD: Agent-type Aware Flow Matching with Entropy-Regularized Distillation for Diverse Traffic Simulation
Seulbin Hwang*, **Kiyoung Om***, Daejung Kim, and Jinhan Lee.
IEEE Robotics and Automation Letters (RA-L), to appear, 2026; to be presented at ICRA 2027. arXiv | Project

Conference Papers & Workshop Proceedings

Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization
Kiyoung Om*, Kyuil Sim*, Taeyoung Yun*, Hyeongyu Kang, and Jinkyoo Park.
NeurIPS (Poster), 2026; NeurIPS SPIGM Workshop (Oral), 2025. Workshop | arXiv | Code

Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function
Hyeongyu Kang*, Jaewoo Lee*, Woocheol Shin*, **Kiyoung Om**, and Jinkyoo Park.
ICLR (Poster), 2026. Paper | arXiv | Code

Diffusion Alignment as Variational Expectation-Maximization
Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, Taeyoung Yun, **Kiyoung Om**, and Jinkyoo Park.
ICLR (Poster), 2026. Paper | arXiv | Code

Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization
Taeyoung Yun*, **Kiyoung Om***, Jaewoo Lee, Sujin Yun, and Jinkyoo Park.
ICML (Poster); FPI @ ICLR Workshop, 2025. Paper | arXiv | Code

Industry White Papers

AI-accelerated Heuristics for Production Planning
Jung, J., Sim, Y., Lee, J. H., **Om, K.**, Shin, H., & Kwon, C.
Samsung Electronics-KAIST industry white paper, 2025.

EXPERIENCE

NAVER LABS, Robotics & Embodied AI Team | Mar 2026 - Jul 2026

Research Intern | Seongnam, South Korea

Researched learning-based traffic simulation for autonomous driving, with a focus on closed-loop covariate shift and realistic multi-agent behavior.

Contributed to Flow-ERD, a flow matching based traffic simulator that ranked 1st on the WOSAC 2025 Sim Agents Challenge leaderboard (July 2026).

PROJECTS

Samsung Electronics-KAIST AI Industry-Academia Collaboration | Mar 2025 - Sep 2025

AI Research Assistant | Daejeon, South Korea

Developed AI-accelerated production planning heuristics, reducing runtime by approximately 75%.

Korea University Solar Cell Lab AI Industry-Academia Collaboration | Apr 2023 - Jul 2023

AI Application Advisor | Seoul, South Korea

Built CNN-based solar-cell reflection models and weather-based experimental planning tools.

TEACHING EXPERIENCE

The University of Texas at Austin | Fall 2026 - Present

Teaching Assistant - ECE 460J Data Science Laboratory | Prof. Amy Zhang | Austin, TX, USA

KAIST Industrial & Systems Engineering | Mar 2024 - Jun 2024

Teaching Assistant - IE343 Statistical Machine Learning | Prof. Jinkyoo Park | Daejeon, South Korea

Korea University | Jul 2023 - Aug 2023

Teaching Assistant - Deep Learning Programming Study | Prof. Sungbin Lim | Seoul, South Korea

AWARDS

Academic Excellence Award | 2022

Korea University

Capstone Excellence Award | 2022

Korea University

Topic: Understanding the Lifecycle and Gap Technologies of Lithium-Ion Batteries

Commander's Commendation | 2020

Republic of Korea Army

Speech Contest on Practicing the Core Values of the Army and Showcasing Examples

EXTRACURRICULAR ACTIVITIES

KUBIG (Korea University BIG data research club) | Jul 2022 - Jul 2023

Regular Member | Seoul, South Korea

Led AI study groups and built projects in vehicle hazard detection and algorithmic trading.